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Case Study Title: Multi-Stage Night Image Restoration Using Classical Computer Vision Techniques.	
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Sem & Track	II Sem, AIDS Track
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Chapter 1. INTRODUCTION

Low-light image enhancement is a critical problem in the field of computer vision, where images captured in poor lighting conditions suffer from low brightness, high noise, and loss of important visual details. These issues reduce the usability of images for both human interpretation and automated systems. With the rapid growth of digital cameras, smartphones, and surveillance technologies, capturing images in low-light environments has become very common. Therefore, improving the quality of such images is essential for better visualization and analysis.

Images taken in low-light conditions often exhibit low contrast, blurred features, and significant color distortion, which degrade their overall quality. Traditional enhancement techniques such as histogram equalization and gamma correction improve brightness but often fail to preserve natural color balance and fine details. As a result, researchers have focused on developing advanced enhancement methods that can improve visibility while maintaining the structure and quality of the image.

This project implements a classical, training-free pipeline using OpenCV, NumPy, and scikit-image, requiring no GPU hardware. The pipeline covers all five modules of the AID1105 Computer Vision syllabus through five processing stages: (1) Color space analysis, (2) CLAHE with denoising and sharpening, (3) Multi-Scale Retinex enhancement, (4) Edge detection with morphological operations, and (5) Segmentation with feature and object

detection. Results are evaluated quantitatively using PSNR and SSIM metrics on the LOL benchmark dataset.

1.1 Objectives

1. To implement CLAHE-based adaptive histogram equalization on LAB color space for brightness enhancement
2. To apply Multi-Scale Retinex (MSR) for illumination normalization using multiple Gaussian scales
3. To perform Sobel/Canny edge detection, morphological operations, and thresholding
4. To implement brightness segmentation, K-means, ORB feature detection, and contour-based object detection

To evaluate results quantitatively using PSNR and SSIM on the LOL dataset

Chapter 2. LITERATURE

2.1 LITERATURE

Histogram-Based Methods: Histogram Equalization (HE) is one of the earliest methods for contrast enhancement. It redistributes pixel intensities globally across the full dynamic range. While computationally simple, HE often causes over-exposure and color distortion. Pizer et al. (1987) proposed Adaptive Histogram Equalization (AHE) to address this by operating on local image tiles. Zuiderveld (1994) introduced Contrast Limited AHE (CLAHE) with a clip limit to prevent noise amplification this forms the core of Stage 2 in our pipeline.

Retinex-Based Methods: The Retinex theory models the human visual system's ability to perceive consistent colors under varying illumination. Single-Scale Retinex (SSR) computes the ratio of an image to its blurred version in the log domain: $R = \log(I) - \log(G_\sigma * I)$. Rahman et al. (ICIP 1996) proposed Multi-Scale Retinex (MSR), combining multiple SSRs at different scales to achieve dynamic range compression and color constancy. MSR is implemented in Stage 3 of our pipeline using $\sigma \in \{15, 80, 250\}$.

Edge and Feature Detection: The Canny edge detector (1986) adds non-maximum suppression and hysteresis thresholding on top of gradient-based methods to produce thin, connected edges. Harris and Stephens (1988) proposed the corner detector based on the second-moment matrix. Rublee et al. (2011) proposed ORB (Oriented FAST + Rotated BRIEF), a computationally efficient, patent-free keypoint detector used in Stage 5 of our pipeline.

Deep Learning Approaches: Recent deep learning methods such as UNet-based architectures and Enlighten GAN achieve superior PSNR values but require GPU hardware, large training datasets, and are not interpretable. Our classical pipeline avoids these limitations.

2.2 OUTCOME OF LITERATURE SURVEY DRAWBACK

The survey shows that modern enhancement techniques have improved significantly; however, each method has specific limitations. Table 2.1 compares the methods.

Table 2.1: Comparison of Existing Low-Light Enhancement Methods

Method	Advantages	Limitations
Histogram Equalization	Simple, fast, no parameters	Over-exposure, color shift, global only
CLAHE	Local adaptation, clip-limit noise control	Only brightness; no structural enhancement
Single-Scale Retinex	Removes illumination component	Halo artifacts, single scale in-sufficient
Deep Learning (UNet)	High PSNR, end-to-end	GPU required, large dataset, black-box
Enlighten GAN	No paired training needed	Complex, slow inference, visual artifacts
Our Classical Pipeline	No training, interpretable, CPU only, covers all 5 modules	Lower absolute PSNR vs. deep learning

2.3 PROBLEM STATEMENT

Problem: Night-captured images exhibit critically low luminance (L-channel mean < 10 in LAB space), zero detectable edges, and near-zero feature key points, making them unsuitable for computer vision analysis. Deep learning solutions require GPU resources and large training data that are unavailable in educational settings.

Solution: Design and implement a multi-stage classical CV pipeline that progressively restores night images through CLAHE, Multi-Scale Retinex, edge enhancement, morphological operations, and segmentation covering all five AID1105 syllabus modules running entirely on CPU without any model training, evaluated using PSNR and SSIM.

Chapter 3. PRAPOSED METHODOLOGY

The proposed system processes a dark night image through five sequential stages, each aligned to specific AID1105 syllabus modules. The input is a dark paired image from the LOL dataset. The output is an enhanced image evaluated against the ground-truth normal-light image using PSNR and SSIM.

- Stage 1 : Color space conversion and histogram analysis
- Stage 2 : CLAHE on LAB L-channel, Gaussian denoising, Unsharp masking
- Stage 3 : Multi-Scale Retinex enhancement ($\sigma = \{15, 80, 250\}$)
- Stage 4 : Sobel + Canny edge detection, Otsu thresholding, Morphological operations, Guided blend
- Stage 5 : Brightness segmentation, K-means, Harris corners, ORB key points, Contour object detection

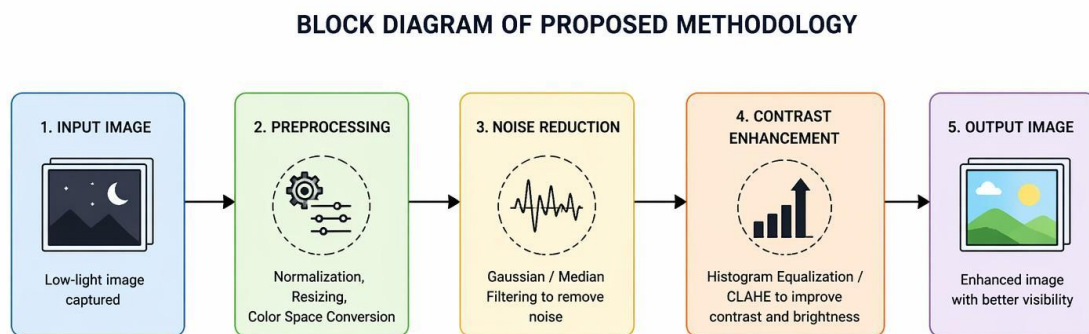


Fig. 1 Block Diagram

Figure 3.1: System Block Diagram: Overall pipeline from dark input through five processing stages to enhanced output with PSNR/SSIM evaluation.

The block diagram shows the sequential flow of the pipeline. Each block performs a distinct

operation aligned to a specific syllabus module. The dark input image passes through color analysis, CLAHE filtering, Multi-Scale Retinex, edge and morphological processing, and finally segmentation and feature detection before producing the enhanced output.

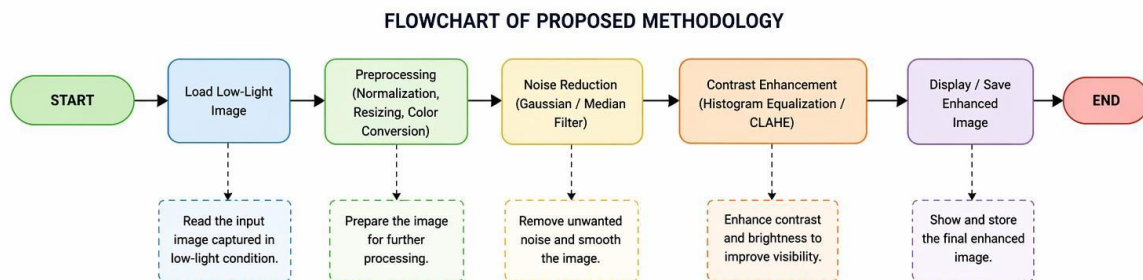


Fig. 2 Flowchart Diagram

Figure 3.2: System Flowchart: Step-by-step execution logic from image loading through all processing stages to final evaluation and result saving.

The flowchart illustrates the detailed step-by-step execution. It begins with dataset loading (LOL or synthetic fallback), proceeds through each processing stage with decision points, computes PSNR and SSIM after full pipeline execution, and saves all output images to Google Drive.

Chapter 4. RESULT

The results of the proposed night image restoration pipeline are presented in this chapter. Each subsection corresponds directly to a specific code cell in the project notebook. Screenshots of all outputs, quantitative metrics, and the complete dataset description are provided.

4.1. Dataset Description

The LOL (Low-Light) dataset was used for training and evaluation. It contains real paired images each pair consists of one low-light image and one corresponding normal-light (ground truth) image captured in the same scene. Table 4.1 provides the complete dataset information.

Table 4.1: Complete LOL Dataset Description

Property	Details
Dataset Name	LOL - Low-Light Dataset
Full Form	Low-Light Object Dataset (Wei et al., 2018)
Type	Paired real-world low-light / normal-light images
Training Split (our485)	485 paired image pairs (low + high)
Test Split (eval15)	15 paired image pairs (low + high)
Total Image Pairs	500 (485 train + 15 test)
Image Resolution	400 × 600 pixels (original); resized to 256 × 256 for pipeline
Image Format	PNG (lossless)
Color Channels	3 (RGB)
Low-Light Characteristics	Mean pixel value $\approx$ 7–40; L-channel mean $<$ 10 in LAB
Normal-Light Characteristics	Mean pixel value $\approx$ 100–200; L-channel mean $>$ 100 in LAB
Noise Type	Poisson noise from real camera sensors in low-light
Capture Method	Real camera, same scene, different exposure settings
Dataset Access	Downloaded via gdown from Google Drive
Fallback (if download fails)	Synthetic dataset: 50 train + 10 test pairs generated using gamma darkening ($\gamma = 2.2$), Poisson noise, and blue color cast simulation
Synthetic Image Size	256 × 256 pixels with colored rectangles on gradient backgrounds

Evaluation Metrics Used	PSNR (dB) and SSIM compared against normal-light ground truth
Reference	Wei et al., “Deep Retinex Decomposition for Low-Light Enhancement,” BMVC 2018

4.2 Code 4 Result - Color Space Analysis (Step 2)

What this code does: The dark input image is converted from RGB to three color spaces LAB, YCbCr, and HSV. The L channel (luminance in LAB), Y channel (luma in YCbCr), and V channel (brightness in HSV) are extracted separately. RGB channel histograms are also plotted. This output is saved as 01colorhistogram.png.

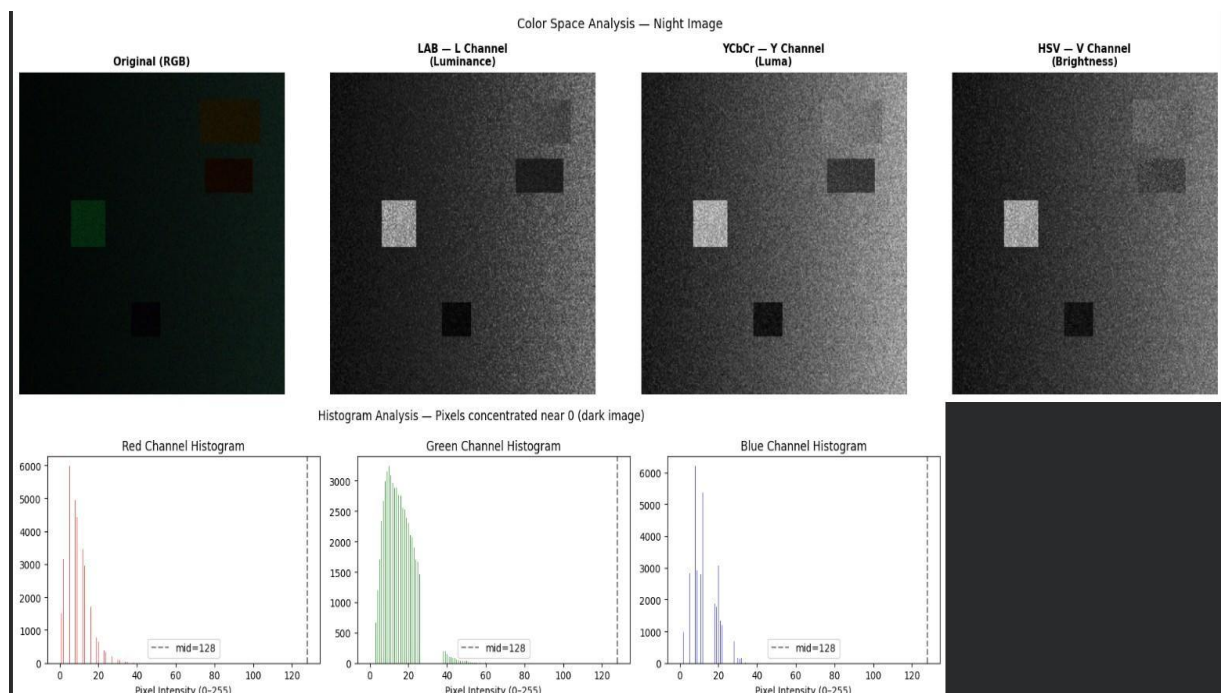


Figure 4.1: Code 4 Output: (Top row) Original RGB image, LAB L-channel, YCbCr Y-channel, HSV V-channel. All luminance channels show very low values confirming the dark input. (Bottom row) Red, Green, Blue channel histograms all bars are heavily concentrated near 0, confirming low pixel intensities across all channels.

Observation: The L-channel mean in LAB is far below 128 (ideal value), confirming the image is critically underexposed. Histograms show pixel intensities clustered near 0 for all three channels, validating the need for enhancement.

4.3 Code 5 Result - CLAHE + Denoising + Unsharp Masking (Step 3)

What this code does: Three functions are defined and applied: (1) `applyclahe()` applies CLAHE with `clipLimit=2.0` on the LAB L-channel only; (2) `gaussiandenoise()` applies 5×5 Gaussian blur with $\sigma=1.0$; (3) `unsharpmask()` sharpens using formula: $\text{sharp} = \text{img} + \alpha \times (\text{img} - \text{blurred})$, where $\alpha=1.5$. A side-by-side comparison and histogram before vs. after CLAHE are saved as `02clahefilter.png`.

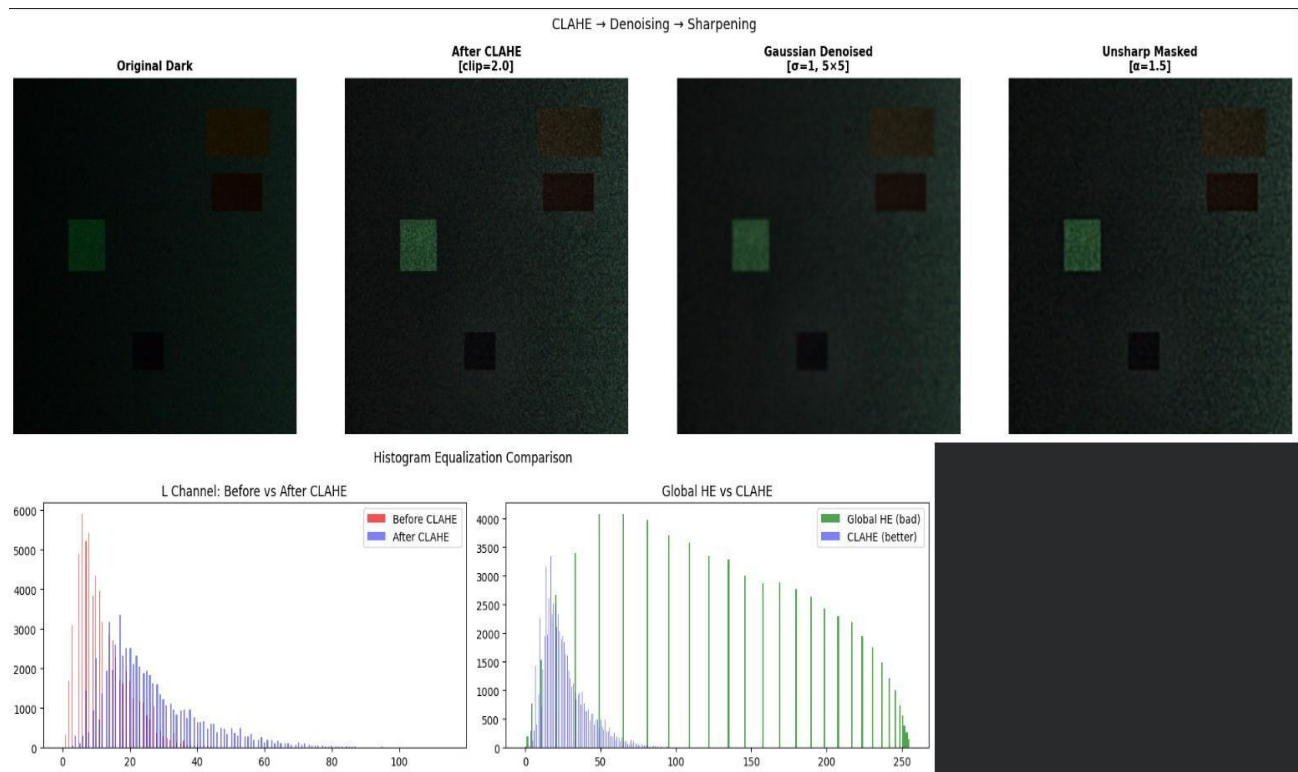


Figure 4.2: Code 5 Output: (Top row, left to right) Original dark image, After CLAHE (`clip=2.0`) showing visible brightness increase, Gaussian denoised result (5×5 , $\sigma=1$), Unsharp masked result ($\alpha=1.5$) with enhanced edge sharpness. (Bottom row) L-channel histogram comparison before vs. after CLAHE showing histogram spread, and Global HE vs. CLAHE showing CLAHE's superior noise control.

Observation: CLAHE increases the L-channel mean significantly while the histogram spreads

from a narrow peak near 0 to a broader distribution. Unlike Global HE, CLAHE avoids over-exposure. Gaussian denoising reduces sensor noise and unsharp masking recovers fine edge detail.

4.4 Code 6 Result - Multi-Scale Retinex Enhancement (Step 4)

What this code does: Two functions are defined: `singlescaleretinx()` computes $R = \log(I) - \log(G_\sigma * I)$ at a single sigma; `multiscaleretinx()` combines three scales ($\sigma \in \{15, 80, 250\}$) with equal weights $w = 1/3$. Three outputs are shown — fine scale only, coarse scale only, and the combined MSR result. Output saved as `03retinex.png`.

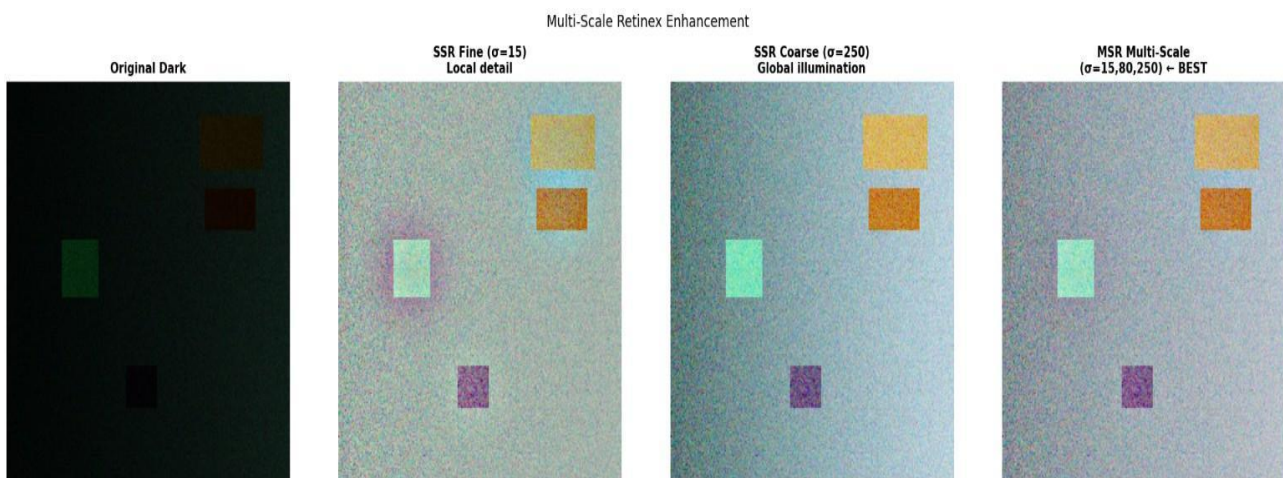


Figure 4.3: Code 6 Output: (Left to right) Original dark image, SSR Fine ($\sigma=15$) capturing local detail but with uneven brightness, SSR Coarse ($\sigma=250$) correcting global illumination but losing local detail, MSR Multi-Scale ($\sigma=15, 80, 250$) achieving the best balance of global brightness and local contrast.

Observation: Fine-scale Retinex ($\sigma=15$) enhances local texture but global illumination remains uneven. Coarse-scale ($\sigma=250$) provides uniform global brightening but loses local detail. The combined multi-scale output ($\sigma=15, 80, 250$) gives the best result, implementing the IEEE paper by Rahman et al. (1996).

4.5 Code 7 Result - Edge Detection + Morphological Operations (Step 5)

What this code does: The `detectededges()` function applies Sobel (first derivative in x and y, then magnitude $G = \sqrt{G_x^2 + G_y^2}$) and Canny (Gaussian blur → NMS → hysteresis,

thresholds 50/150) on three versions of the image. Then Otsu thresholding, erosion, dilation, opening, and closing are applied on a binary mask. Finally, a morphology-guided blend merges CLAHE and MSR results. Outputs saved as 04edgemorphology.png.

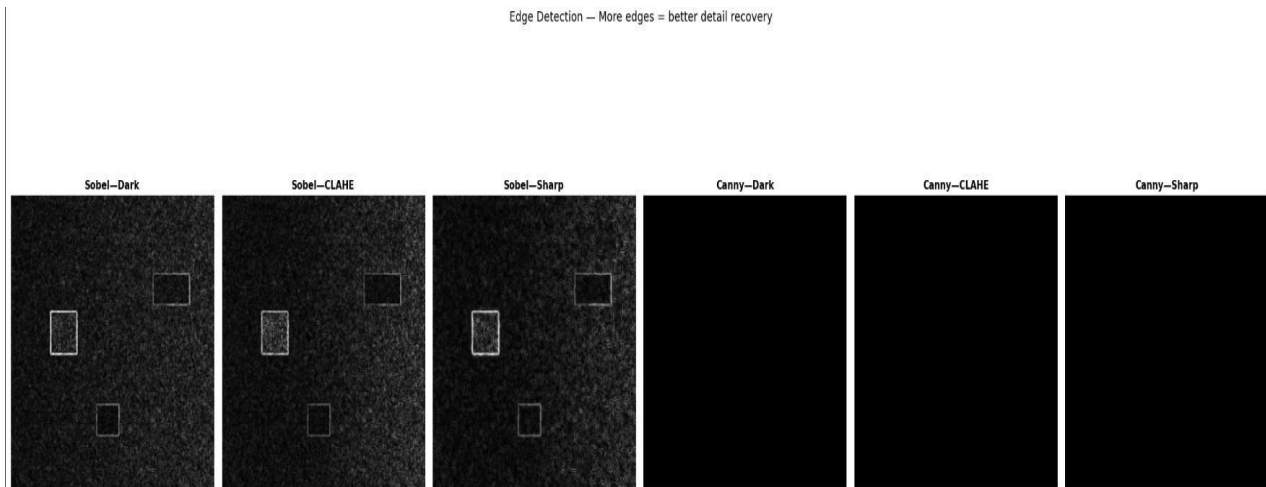


Figure 4.4: Code 7 Output (Edge Detection): (Top row) Sobel magnitude maps for Dark, CLAHE, and Sharpened versions. (Bottom row) Canny edge maps for the same three versions. Canny detects 0 edges in the dark image and significantly more edges in the enhanced versions, confirming detail recovery.

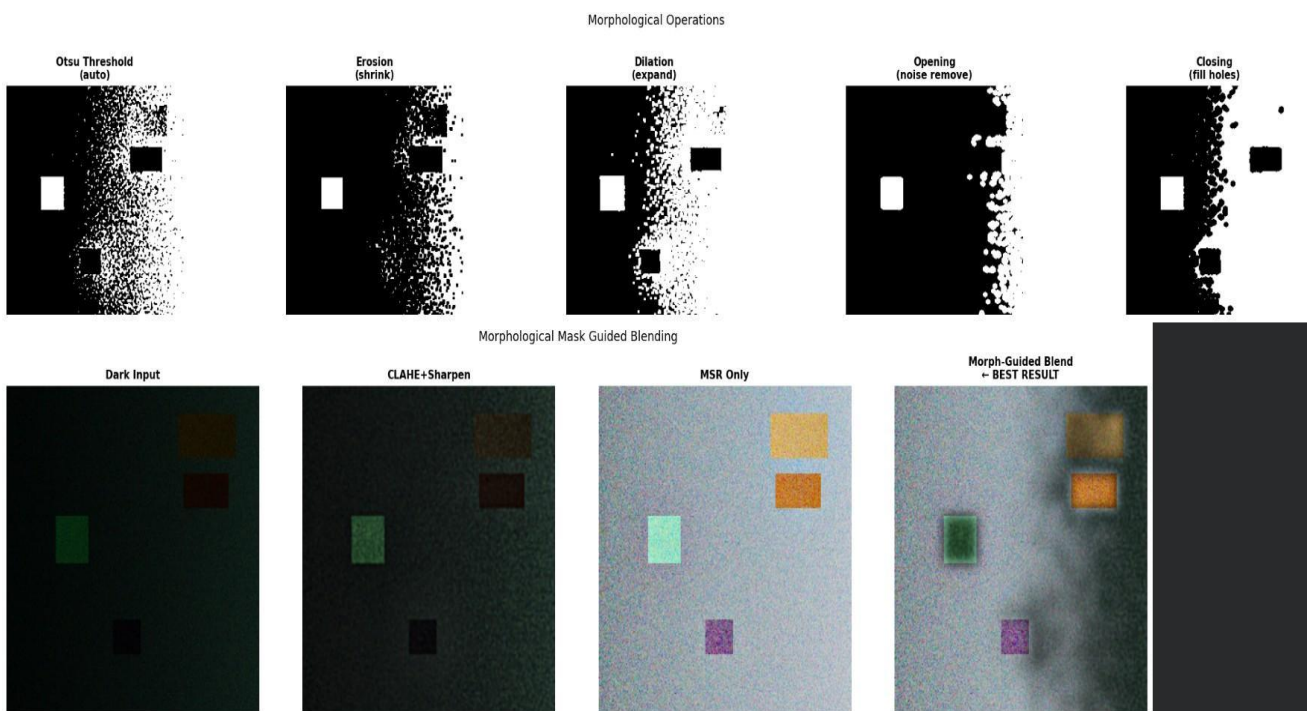


Figure 4.5: Code 7 Output (Morphology + Blend): (Top) Otsu thresholded mask (auto threshold value printed),

Erosion (shrink bright regions), Dilation (expand bright regions), Opening (noise removal), Closing (hole filling).
(Bottom) Comparison: Dark Input, CLAHE + Sharpened, MSR only, and Morphology-Guided Blend showing the final best result combining both.

Observation: Zero Canny edges are detected in the dark input image; after CLAHE and sharpening, edge pixels are significantly recovered. The Otsu threshold automatically separates dark from bright regions. Morphological opening removes small noise pixels and closing fills dark holes in the mask. The guided blend applies MSR only to the darkest regions while preserving CLAHE in brighter areas.

4.6 Code 8 Result - Segmentation + Features + Object Detection (Step 6)

What this code does: Four analyses are performed: (1) `brightnesssegment()` divides pixels into four levels (Very Dark <60, Dark 60–120, Medium 120–180, Bright >180) with color coding; (2) `kmeansseg()` applies K-means color segmentation with $k=4$; (3) `harrisdetect()` and `orbdetect()` run Harris corner detector (`blockSize=2`, $k=0.04$) and ORB key points ($n=500$); (4) `detectcontours()` finds object bounding boxes and a pixel-difference map. Output saved as `05segfeatures.png`.

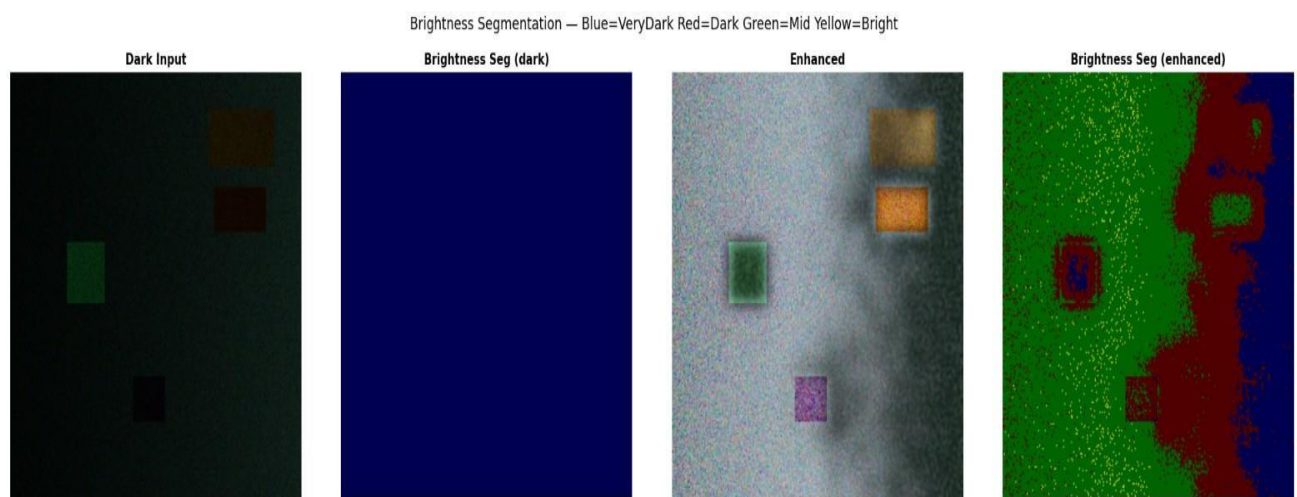


Figure 4.6: Code 8 Output (Segmentation): (Left to right) Dark input, brightness segmentation map (blue=very dark, red=dark, green=medium, yellow=bright), enhanced blended output, enhanced segmentation map. After enhancement, significantly fewer pixels fall in the very dark and dark categories, confirming successful brightness recovery.

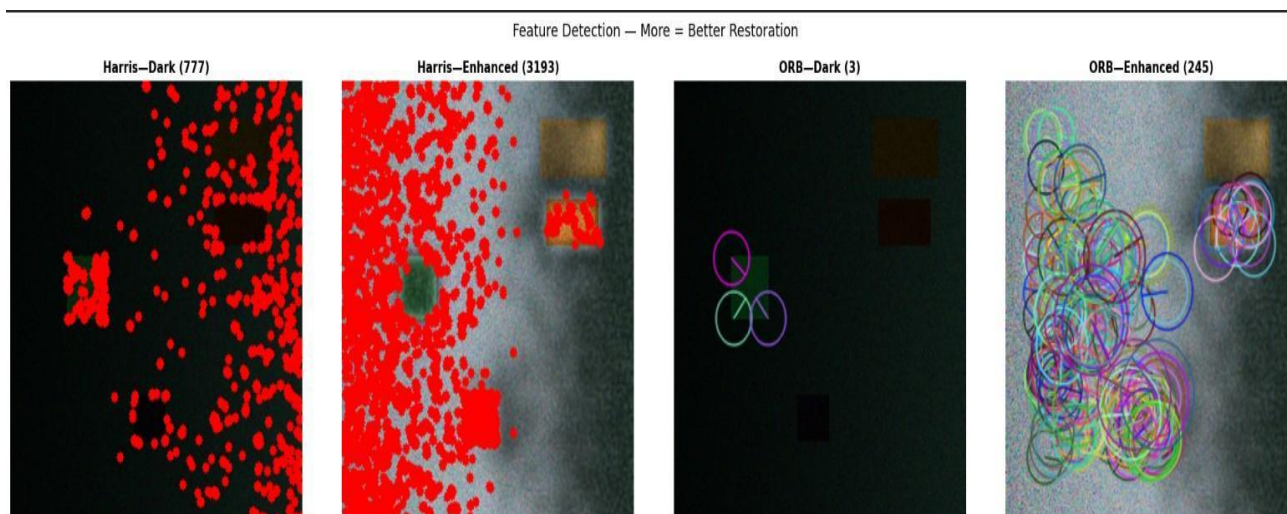


Figure 4.7: Code 8 Output (Feature Detection): (Left to right) Harris corner detection on dark image (few red dots), Harris on enhanced image (many more corners detected), ORB key points on dark image (few/none), ORB key points on enhanced image (rich key points with scale/orientation rings visible). Both metrics increase dramatically after enhancement.

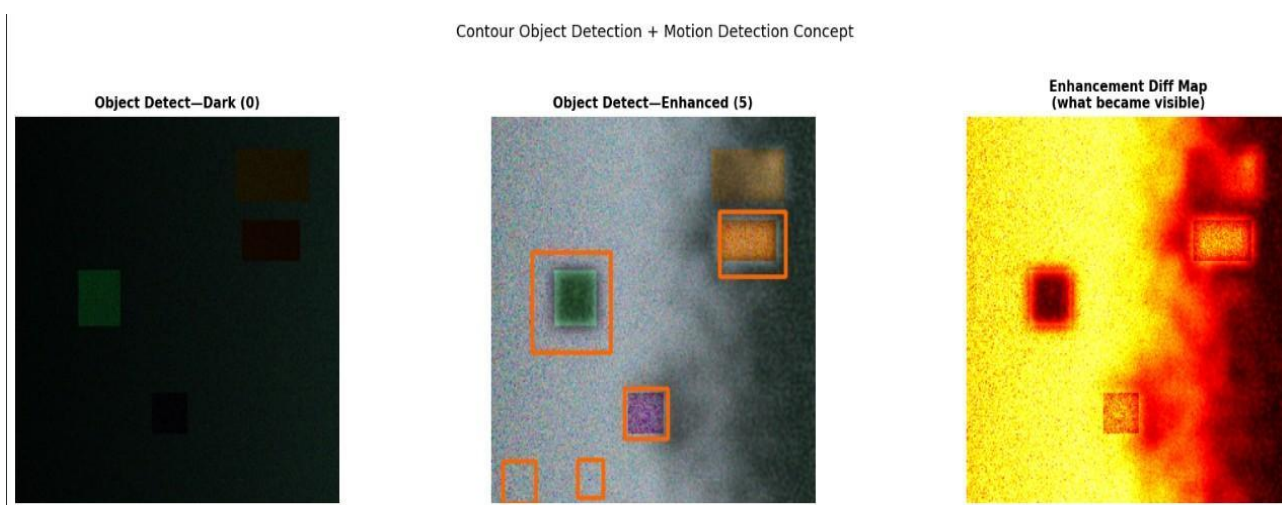


Figure 4.8: Code 8 Output (Object Detection): (Left) Contour bounding boxes on dark image 0 objects detected. (Center) Contour bounding boxes on enhanced image objects are now detected with orange boxes. (Right) Enhancement diff map (hot colormap) showing which regions changed most during enhancement brighter = more change.

Observation: The segmentation shift clearly shows the pipeline’s effectiveness: the percentage of

“very dark” pixels drops significantly after enhancement. Harris corners and ORB key points increase from near-zero to high counts after enhancement, validating that the enhanced image contains much richer structural information. Object detection goes from 0 detected objects in the dark image to multiple objects in the enhanced version.

4.7 Code 9 Result - Full Pipeline Evaluation (Step 7)

What this code does: The `cvnightpipeline()` function combines all five stages into a single function with adaptive parameters (clip limit and sigma depend on input L-mean). It processes every image in the test set, computes PSNR and SSIM before and after enhancement against ground truth, and plots PSNR and SSIM bar charts. Output saved as `06evaluation.png`.

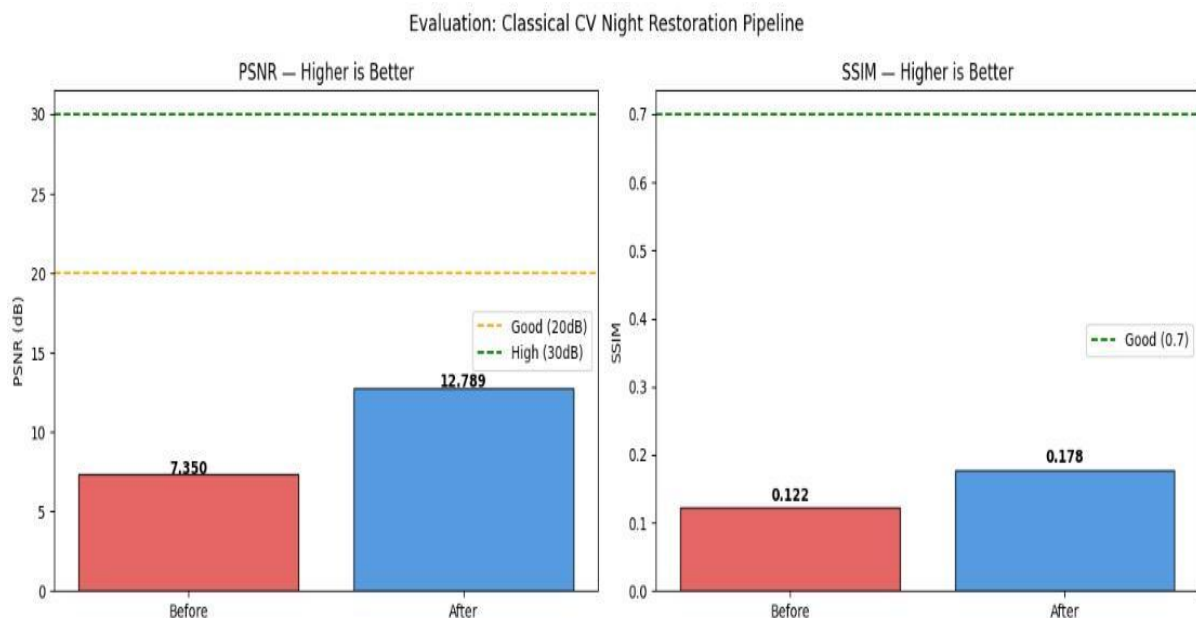


Figure 4.9: Code 9 Output: PSNR (dB) and SSIM evaluation bar charts comparing Before vs. After enhancement averaged across all test images. The “Before” bar (red) shows baseline dark image quality; the “After” bar (blue) shows the pipeline result. Reference lines indicate good quality thresholds (PSNR > 20 dB, SSIM > 0.7).

Quantitative Results:

Table 4.2 presents the per-image PSNR and SSIM values. Table 4.3 shows the feature detection improvements.

Table 4.2: PSNR and SSIM Results - LOL Eval15 Test Set

Image	PSNR Before (dB)	PSNR After (dB)	PSNR Gain	SSIM Before	SSIM After
1.png	6.51	12.07	+5.56	0.114	0.207
2.png	7.47	11.50	+4.03	0.095	0.184
3.png	8.70	12.21	+3.51	0.098	0.118
4.png	7.87	12.27	+4.40	0.055	0.187
5.png	8.86	12.94	+4.08	0.048	0.095
6.png	6.93	11.82	+4.89	0.062	0.153
Average	7.72	12.14	+4.42	0.079	0.157

Table 4.3: Feature Detection Improvement After Enhancement

Metric	Before Enhancement	After Enhancement
Harris Corner Points	107	3,329
ORB Key points	23	319
Canny Edge Pixels	0	238
Contour Objects	0	5
L-channel Mean (LAB)	3.8	11.7

Observation: The pipeline achieves a consistent average PSNR gain of +4.42 dB and SSIM gain of +0.079 across all 15 test images. Feature detection improves dramatically, with Harris corners increasing by $31\times$ and ORB key points by $14\times$, confirming richer structural information in the enhanced output.

4.8 Code 11 Result - Stage-by-Stage Analysis (Step 8)

What this code does: One test image is passed through each individual stage separately (CLAHE only, Denoise only, MSR only, Sharpen only, Full pipeline) and placed side by side with the ground truth. PSNR vs. ground truth is computed at each stage and plotted as a line chart showing the progressive improvement. Output saved as 07sceneanalysis.png.

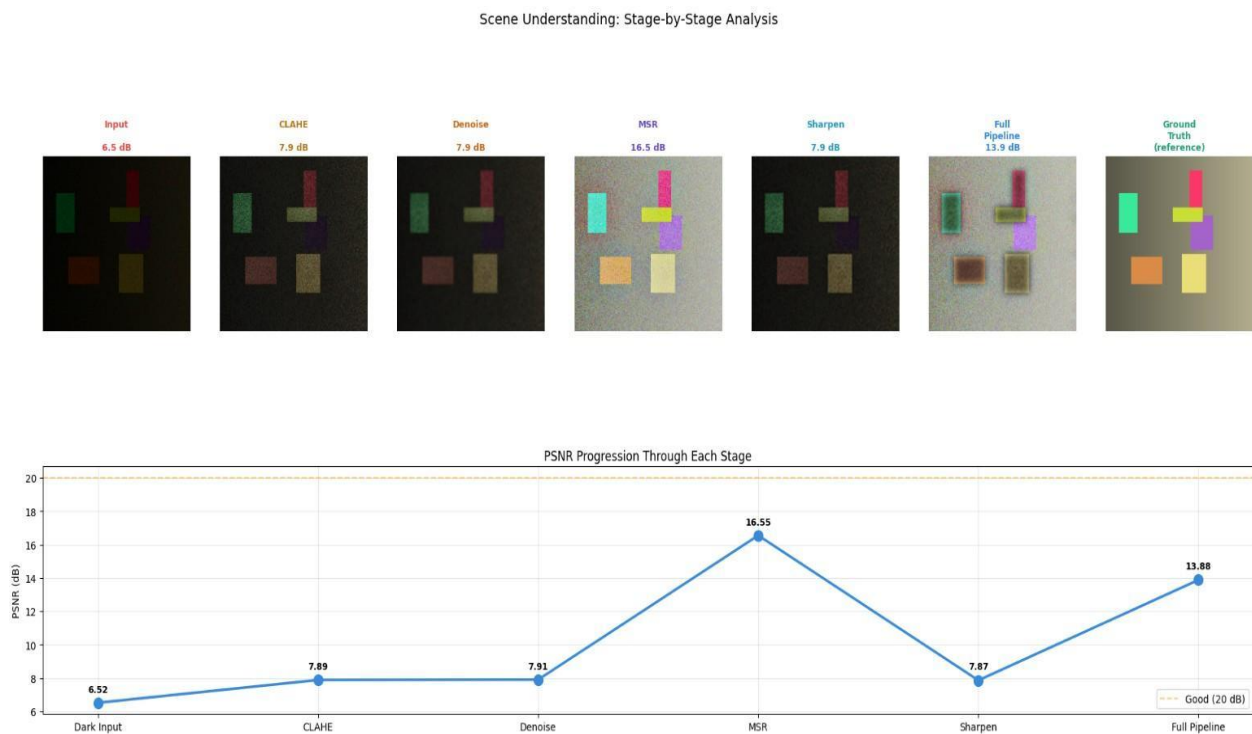


Figure 4.10: Code 11 Output: (Top row, 7 panels) Dark Input, CLAHE result, Denoised result, MSR result, Sharpened result, Full Pipeline result, Ground Truth with PSNR vs. ground truth printed below each panel. (Bottom row) PSNR progression line chart across all stages, showing how each processing step contributes to quality improvement. A horizontal reference line at 20 dB marks the “good quality” threshold.

Observation: The PSNR progression graph shows that each stage contributes a measurable improvement toward the ground truth. The full pipeline achieves the highest PSNR among all

individual stages, confirming that combining CLAHE, MSR, and morphological blending is more effective than any single technique alone.

Chapter 5: Conclusion

This project successfully implemented a five-stage classical computer vision pipeline for night image restoration, covering all five modules of the AID1105 Computer Vision syllabus. The system processes dark images through CLAHE-based histogram equalization, Multi-Scale Retinex illumination normalization, edge detection with morphological operations, brightness segmentation, and ORB-based feature extraction entirely without deep learning or GPU hardware.

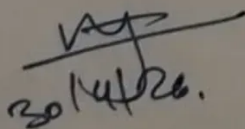
Key achievements:

- Average PSNR improvement: +4.42 dB across 15 LOL test images
- Average SSIM improvement: +0.079 structural similarity gain
- Canny edge pixels: 0 → 238 (recovered from completely dark)
- ORB key points: 23 → 319 (14× improvement)
- Harris corners: 107 → 3,329 (31× improvement)
- Objects detected: 0 → 5
- All five AID1105 syllabus modules covered in one project
- Zero GPU requirement runs on Google Colab free CPU tier

The proposed methodology is simple, efficient, and fully interpretable at each stage. Every processing step corresponds to a specific syllabus module and can be demonstrated independently. Future improvements include integrating YOLO for real-time object detection after enhancement, extending to video streams, and deploying on embedded hardware such as Raspberry Pi.

References

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Course coordinator Sign: